

# What Makes A Good Reference Manager?

## Quantitative Analysis of Bibliography Management Applications

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Reference managers have been widely used by researchers and students. While previous performed qualitative analysis for reference managers, it is unclear how to assess these tools quantitatively. In this paper, we attempted to quantify the physical and mental effort to use a reference manager. Specifically, we use a keystroke and mouse move logger, RUI, to record and analyze the user's activities and approximate the physical and mental effort. We also use pre- and post-study surveys to keep track of the participant's preferences and experiences with reference managers, and also their self-reported task load (NASA TLX Index.) In this pilot work, we first collected 69 pre-study surveys from graduate students to understand their experience with reference managers, and then conducted user study with 12 voluntary participants. Four common reference managers, Mendeley, Zotero, EndNote, and RefWorks, were included in our study. The results show, for the same task, different software might require different levels of effort, and users generally prefer the tools that require less effort. We also observe that although these reference managers share similar features, the differences in their presentation and organization matter. Factors such as pricing, cloud sync and accuracy of bibliography generation also influence the preference of users. We conclude this work by providing a set of guidelines for users and developers.

CCS Concepts: • **Human-centered computing** → **Usability testing**; *Activity centered design*.

Additional Key Words and Phrases: reference managers, task analysis, mental/physical efforts

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## 1 INTRODUCTION

Reference managers, or reference management applications, are computer applications or systems that help users to collect, manage and format references and citations for academic purposes. It is both time-consuming and tedious to manually collect and keep track of all citations, but with the help of reference managers, users will feel much more comfortable and relieved. Reference managers are optimized and well used for writing literature reviews [10], and can help to make PDF files of the paper cited to be instantly viewable and searchable [11].

Yet it is not always agreed which reference manager is the "best". The website G2 shows that the most liked reference manager is Mendeley<sup>1</sup> with 170 reviews and 4.3/5 rating. On another website, Scribendi, RefWorks is the top pick from the editor.<sup>2</sup> On the other hand, no outstanding or novel product has been seen in this domain. The history of publicly available, web-based reference management applications started decades ago, but the most-used reference managers like Mendeley and EndNote are still old-fashioned – they require local distribution of the software and are Operating

<sup>1</sup><https://www.g2.com/categories/reference-management>

<sup>2</sup>[https://www.scribendi.com/advice/reference\\_management\\_software\\_solutions.en.html](https://www.scribendi.com/advice/reference_management_software_solutions.en.html)

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System-specific. These reference managers are generally much goal-oriented and have simple logic, which is easier to start with but lacking considerations of usability [11]. Some recent updates with the reference managers lie in the availability of personal online accounts that makes better cross-device experiences, but the core functionality remains unchanged — even if non-efficient — due to the fact that the majority of users are used to it. Thus, the development of reference managers is rather slow.

For this study, we address this problem by conducting a quantitative, comparative study among four commonly used reference managers – Mendeley, Zotero, EndNote, and RefWorks, aiming to distinguish different level of efforts for users to operate them, and associate the efforts to the user’s preference. Specifically, we will reference and improve the ideas in previous works and conduct a mixed analysis of user efforts by 1) recruiting participants that have exposure to reference managers and asking for their preference, 2) qualitatively comparing the functionalities and features of each reference management application, 3) recording and quantitatively analyzing the mouse movements and keystrokes in operating the application, 4) building separate Key-stroke Level Models (KLM) [4] with Cogulator for each of the applications, 5) using NASA TLX index [8] to further estimate the workload for each participant and 6) finding the association between the workload, efforts and features with the participants’ preference.

The major contributions of our work lie in the following two aspects:

- (1) We explore a novel, objective evaluation scheme for the reference managers that incorporate both qualitative evaluation of the software’s features and functionalities and quantitative evaluation of a user’s effort during a routine task, and associate the result with the user’s preference. The scheme is believed to be valuable to similar usability analysis or software evaluation tasks.
- (2) Based on the results we obtained from the study, we provide suggestions to users who are going to choose their preferred reference manager, and we also draw the guidelines for developers in making a better reference manager from three aspects: correctness, effort and functionalities. We believe the guidelines will help developers to improve the current reference managers and benefit the users.

## 2 RELATED WORK

### 2.1 Comparative Studies of Reference Managers

Previous works in various forms have been seen to compare different reference managers. The majority of them conducted surveys to compare between the users’ opinions towards different reference managers. [10] conducts a survey on reference manager users about their usability and emphasizes the ease-of-use issues concerning the integration of the software with other programs and the sharing of reference databases among researchers. Some work also compare the functional features for different reference managers, like device compatibility, system requirements, working logics, and prices[14]. Similar idea was adopted in [15] that compares EndNote, Zotero, Connotea, and Mendeley, where each tool is analyzed in terms of their features in accessing, collecting, organizing, collaborating, and citing/formatting. In the work presented by Hensley[9], RefWorks, EndNote, and Zotero are compared in terms of their benefits and drawbacks from a librarian’s perspective. Other works create evaluation metrics like error rate in importing/exporting [6] and average importing time for one bibliography entry [13]. [2] quantified and compared the fields import by Google Scholar on RefWorks, Mendeley, and EndNote, and further visualized them with radar plots. However, few work considered the actual amount of effort for users to complete a routine task on reference managers, making their analysis less grounded and more subjective.

## 2.2 Usability Evaluation of Softwares

The comparison of reference managers can take relevant idea from the assessment of usability. Various software quality models have been proposed to improve the usability of software [1]. There are more concrete methods in evaluating the usability of software. [12] proposes that software ergonomics can have a quantitative basis with regard to multiple criteria of usability concern in comparing interfaces. A first prescreening phase takes expert judgments and the second evaluation phase involves user-based assessment. [3] introduces a CogTool that analyzes tasks with an interactive system and predicts the time an expert will take to perform those tasks. An application of such a two-stage idea is presented in [5], where a pre-study survey was conducted before a quantitative study on the usability of three popular two-factor authentication solutions. For the purpose of our study, we would like to ground our reasoning analysis based on quantitative study that measures the user efforts to compare the usability of different reference managers.

## 3 PRE-STUDY SURVEY

To motivate our analysis on reference managers, we first conducted a pre-study survey<sup>3</sup>. Respondents were recruited using snowball sampling<sup>4</sup>. The aims to act as a qualitative pre-study to the user study that follows, identifying the use cases and the potential issues for current reference managers. We ask a set of demographic questions (levels of study, experiences with reference managers) and only include those who had experiences with reference managers before.

### 3.1 Survey Results & Discussion

**3.1.1 Demographics.** A total of 69 completed our pre-study survey. Majority of the participants are graduate students (29% master and 56.5% Ph.D. ). 60.9% of the participants have experiences with reference managers.

**3.1.2 Use Cases.** By analyzing a free-form answers of use cases, we find that common use cases of reference managers including research purposes, class project, paper organization, citation organization, reading papers, and sharing papers. Majority (72.5%) of the responses are for research purpose.

**3.1.3 Previous Experience of Reference Managers.** Common reference managers including Mendeley, EndNote, Zotero, Refworks and EasyBib. The top 4 reference managers that people prefer most is Mendeley (37.5%), Zotero (27.5%), EndNote (20%) and Refworks (5%). As for the usage frequency, 40.5% of them use several times a week and 16.7% of them use everyday. Our survey results also shows that people are satisfied with their current reference managers. They think their preferred reference manager is fast, efficient, and easy to use without hassles, as shown in Figure 1(b). We also show the functionalities that people care in Figure 1(c).

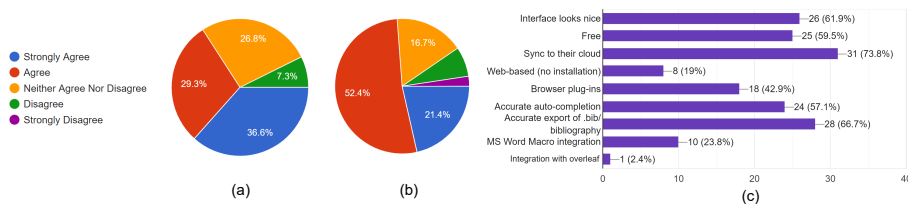


Fig. 1. Pre-study survey: (a) How much do you agree with this statement: “My preferred reference manager is fast and efficient.” (b) How much do you agree with this statement: “My preferred reference manager is easy to use without hassles.” (c) How do you tell if a reference manager is good? Select all that apply.

<sup>3</sup>The full questionnaire can be found in the Supplementary Material.

<sup>4</sup>We emailed the survey to direct contacts and invited them to pass the survey on to friends. Responses are from students of STEM majors at US universities

## 4 PHYSICAL & MENTAL EFFORT ANALYSIS

In order to quantitatively investigate the efficiency of the chosen reference managers, we conduct a user study to analyze the physical and mental effort from the users when using those applications to perform a routine reference management task. The study went through an ethical review and was IRB-approved.

### 4.1 Investigated Reference Managers

We first review the reference management applications we're studying: Mendeley, Zotero, EndNote and RefWorks.<sup>5</sup>

- **Mendeley** is a free reference manager by Elsevier. In 2018, it was estimated that there are already 5 million users for Mendeley<sup>6</sup>. Some issues regarding Mendeley are the relatively weak auto-completion of information. For newer conference papers or preprint papers, Mendeley constantly fails to find the correct information from its database and result in errors.
- **Zotero** is also a free, open-source reference management application by the Corporation for Digital Scholarship. It has similar workflow logic as Mendeley but is better at importing and indexing PDF inputs. Zotero can achieve very good accuracy and efficiency in creating new entries of reference. Users can also have personal accounts for sync and managing references.
- **EndNote** is a commercial, closed-source reference management application by Clarivate Analytics. EndNote has its own format of library users can use to import/export. It also has a "database search" integration for full version. It has an integration with Google Scholar and reference can be imported in one-click.
- **RefWorks** is a web-based reference management application produced by ProQuest. RefWorks requires institutional subscription. RefWorks allows a strong auto-completion with only the title information of the paper and does not require any installation. Also, the BibTeX export of RefWorks can be directly copied to clipboard, which is very efficient for  $\text{\LaTeX}$  users.

### 4.2 Physical Effort Analysis with RUI

**4.2.1 Design of tasks.** For the analysis of physical effort of users, we design and perform the following tasks that emulate when a user is making a bibliography list from a set of titles. The general guideline formulates as follows:<sup>7</sup>

**GOAL:** The subject is given a list of titles and the corresponding PDF files, without elaborated author or publisher information. (S)he is required to give a full bibliography management list in a bib format with accurate information.

- (1) The subject will first drag or import the PDF files to the library and then works on the reference manager's auto-complete function by only fill in the title of the paper. (This method work for Mendeley & RefWorks)
- (2) After obtaining the references, the subject is also responsible to check that the information provided is accurate. If not, she will have to make corresponding changes for the entries (date format/journal name/author name).
- (3) The subject then output a bibliography list of papers in a .bib file including all the references.

Each subject is asked to follow the required task and operate the applications on a laptop running Windows 10 while being recorded by the RUI. For calculation of mouse movement, we simply adopted a sum of Euclidean distances between each recorded state that are labeled as "Move". Also, the total number of mouse clicks are presented. We try to eliminate bias by using a keyboard shortcut to navigate between Chrome and reference manager applications.

**4.2.2 Result & Discussion.** In this task, a total number of 12 subjects who had previous experience with one or more reference managers are recruited and performed the same task as illustrated above. We present the results of the RUI for different reference managers in Table 1. Each cell contains the averaged statistics (mean) we obtained through the recordings of RUI, and we also calculate the corresponding variance for each statistics. Overall, Zotero shows

<sup>5</sup>In our work, we investigate the stable release of Mendeley 1.19.4, Zotero 5.0.94, EndNote X9.2 and the web-version of RefWorks

<sup>6</sup><https://www.elsevier.com/connect/ten-years-of-mendeley-and-whats-next>

<sup>7</sup>For each of the reference management application, we specify different detailed instructions to emulate an experienced user. Full instructions for each software can be found in the Supplementary Material.

significantly better performance for mouse distance and elapsed time, while RefWorks is the best w.r.t. number of mouse clicks. Both Zotero and RefWorks is efficient in terms of auto-indexing the PDF files with accurate information extraction. In the mean time, RefWorks also provide a convenience online database for searching automatically with the paper title extracted from PDF. Refworks requires slightly more time than Zotero because all PDF files will be uploading to the cloud and the process takes time. Mendeley provides a mediocre performance for all three measures. Interestingly, EndNote performs the worst in the given task. The reason is that EndNote does not automatically extract information from uploaded PDFs. Instead, it requires the users to manually modify the information in our task.

Table 1. The results of the RUI for different reference managers. We measure the physical efforts by Mouse distance(MD, 1e4 Pixels), number of mouse clicks (MC, clicks), and elapsed time (ET, Seconds) respectively. Lower value indicates a better efficiency.

|          | MD                 | MC                   | ET                    |
|----------|--------------------|----------------------|-----------------------|
| Mendeley | 8.55 ± 3.00        | 109.33 ± 36.98       | 325.96 ± 77.03        |
| Zotero   | <b>5.76 ± 2.63</b> | 67.25 ± 26.62        | <b>251.21 ± 92.17</b> |
| EndNote  | 11.25 ± 2.74       | 138.50 ± 37.89       | 408.12 ± 44.67        |
| RefWorks | 6.58 ± 2.24        | <b>62.45 ± 20.02</b> | 294.81 ± 37.92        |

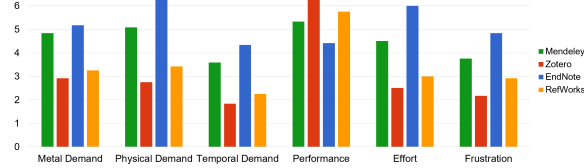


Fig. 2. Participants' ratings on the NASA TLX workload index. Ratings were on a 1-7 scale. Lower values correspond to less demand, effort and frustration. For performance, the higher the better. In general, Zotero and RefWorks have lower workload demand with better performance.

### 4.3 NASA TLX

In addition to the performance measures with RUI, we also asked participants to evaluate the workload of each reference managers using the NASA TLX 7-point Likert scale across six categories [7, 8]. During the user study, we asked the participants to rate NASA TLX index based on their experience after each investigated reference managers. Fig. 2 summarizes the results. We averaged all ratings for 12 users across six categories for each reference managers respectively. For all six aspects, the order is consistent, with Zotero performing the best, RefWorks the second, Mendeley the third, and EndNote the last. It indicates that given our specified task, the reference manager is efficient in terms of the aforementioned order. The reason is the automatic indexing PDF feature for Zotero and RefWorks. In the experiments, we found Mendeley constantly fails to extract accurate information from PDFs thus manual correction is required for bib entries. EndNote does not support information extraction from PDFs so it is most demanding in our given task.

### 4.4 Mental Effort Analysis with Cogulator

The Cogulator is an open-source application for human performance calculation. The Cogulator provides a series of basic operators for Task Analysis, and it takes a natural language syntax. Each of the operators has a default execution time, and the Cogulator allows users to apply different integration of the operators. The output of the Cogulator is quantified as time for the task to be completed. Specifically for KLM, there are operators including click, keystroke, point, and also look, think, verify. We use the default values of time taken for those operators as defined in Cogulator. The operators, corresponding execution time and scripts can be found in the Supplementary material.

Table 2. The results of the Cogulator for different reference managers

|                          | Mendeley | Zotero | EndNote | RefWorks    |
|--------------------------|----------|--------|---------|-------------|
| Total Execution time (s) | 37.8     | 39.9   | 39.1    | <b>29.8</b> |
| Number of Memory chunks  | 50       | 53     | 51      | <b>41</b>   |

*Results and discussions.* As shown in table 2, RefWorks achieves the most efficient performance in completing the task, with the least total execution time and number of operators involved. The performance gain for RefWorks is due to its convenient web plug-in. In our modeling, Mendeley, Zotero, and EndNote take similar longer time and

larger memory chunks. The procedure for those three reference managers are similar<sup>8</sup> except some slightly different movements in execution. Note that the conclusions are different from NASA TLX analysis because in the Cogulator, we did not consider the efforts for correcting wrong information from automatic indexing PDFs from users.

#### 4.5 Retrospective Survey

For participants in the user study, we conducted an exit survey to better understand their experiences with all four reference managers. Interestingly, after experiences with all four reference managers, more than a half of the participants (58.3%) changed their mind in terms of their favorite reference manager from the pre-study survey. As shown in Fig. 3, in pre-study survey, those participants prefer EndNote and Mendeley. While in retrospective survey, they prefer Zotero and RefWorks. Using their preferred reference managers, it would take approximately 1-2 minutes to add one new citation. In addition, 33% of the participants think current reference manager is slow and should be improved while only 16.7% of them disagree. 41.7% of the participants think they are slow but okay. Others (8.3%) indicate there are other issues than efficiency. We calculate the correlation coefficients between 6 NASA indices and 3 RUI measurements and found very high correlations between the variables (Fig. 4), which supports the validity of our user effort measurement approach.

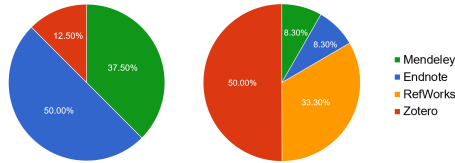


Fig. 3. Users' preferred reference manager before (left) and after the user study (right).

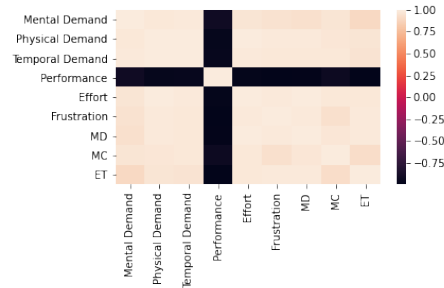


Fig. 4. The covariance matrix heatmap between measures (NASA TLX indices and RUI results). Note that the performance has high negative correlation with others as per the definition.

## 5 IMPLICATIONS & GUIDELINES

Based on our study, we would like to provide suggestions to users on choosing the suitable reference manager. We also propose guidelines to developers on how to improve the efficiency of the current reference managers.

### 5.1 Suggestions to users

We listed and summarized all the important features for the nowadays commonly used reference managers in Table 3 with 5 main categories of the functionalities: (1) Import file; (2) Database connectivity; (3) Complete and correct basic information; (4) Export bibliography; (5) Operating system support. All of the reference managers show complete support for import file formats. Zotero's diverse database connection and strong PDF indexing ability results in fast and accurate performance in our user study, which will suit users who has large amount of PDFs to import. If in more cases, the user only have titles available in hand, RefWorks can work elegantly with the auto-completion, while maintaining way fewer errors and mismatches than Mendeley. Both Mendeley and Zotero will support group sharing of library with

<sup>8</sup>We attache the Cogulator script in the Supplementary Materials.

email invitation, but RefWorks will need organization authorization, which is not the most ideal tool when collaborating. If “Free” is a must, EndNote and RefWorks might not be the choice. All of the four applications support Windows and OS X, but only Mendeley and Zotero support Linux. As for mobile applications, Mendeley and EndNote provide iOS App. Only Mendeley provides an app for Android. For mobile users or cross-platform users, if RefWorks is available, it is highly recommended. A decent portion of our participants admitted that they mainly use the reference managers to storage their papers and work as PDF readers. These users tend to prefer local software than web-based ones.

Besides the reference managers themselves, other tricks that users may adopt to improve their efficiency include (but not limited to) the Chrome plug-ins with Zotero and Mendeley, Google Scholar one-click import with all four software and citation integration provided by major publishers on the paper webpage. These tricks will help import new reference entries more accurately, efficiently and hassle-free.

## 5.2 Guidelines to developers

Based on the study, we propose guidelines to the developers of reference managers from three major aspects: i) correctness, ii) user efforts and iii) functionalities. The three aspects are inter-connected and may be constantly referred to when developing the reference managers.

**5.2.1 Correctness.** From our user study and retrospective post-study survey, the main issues that the participants identified was the errors and mismatches occurred during the import process. Such correctness issue greatly slow down the process and harm the user’s trust in that tool. Also, the auto-completion and online search functions sometimes redirect the paper to a prior version or another paper with very similar names, which will end up requiring users to start over and manually fix all the details. Developers should always put the correctness in the first place when developing the PDF indexing and matching protocols. Challenges to tackle include the citation of pre-print papers, papers with same title but by different authors, identify conference proceedings from journal papers, etc. Besides keeping their back-end querying sources up-to-date, if the algorithm reports low confidence in the result, it should ask the users to step in and make sure the details are correct.

**5.2.2 User efforts.** The results of RUI and retrospective surveys show that users tend to prefer the application with the least physical effort to operate (Zotero) for the specific task. Such discrepancy in effort results from accuracy of importing and ease of correction. While step-by-step instructions and options can benefit new users, experienced users will love to achieve the goal within minimal steps and efforts. Instead of providing all the options in the right-click drop list, simplified ones will help users to quickly navigate and find what he wants. Features like “one-click” export will also make it more efficient for experienced users who know what he’s doing. It is also a good idea to keep all the process in one interface, as switching between different interfaces will increase the mental efforts and slower the process.

| Task                                 | Mendeley | Zotero | EndNote | RefWorks |
|--------------------------------------|----------|--------|---------|----------|
| Import File                          |          |        |         |          |
| Manually Add                         | •        | •      | •       | •        |
| Add Entry from PDF                   | •        | •      | •       | •        |
| Import from .bib/.RIS etc            | •        | •      | •       | •        |
| *Browser Plug-in                     | •        | •      | •       | •        |
| Database Connectivity                |          |        |         |          |
| ArXiv                                | •        | •      | •       |          |
| CiteSeer                             | •        | •      | •       |          |
| IEEE Xplore                          |          | •      | •       |          |
| PubMed                               | •        | •      | •       | •        |
| Complete & Correct Basic Information |          |        |         |          |
| Manually Change                      | •        | •      | •       | •        |
| Online Detail Filling                | •        |        |         | •        |
| PDF information extraction           | •        | •      |         | •        |
| Export Bibliography                  |          |        |         |          |
| Different formats                    | •        | •      | •       | •        |
| Export .bib                          | •        | •      | •       | •        |
| *MS Word Integration                 | •        | •      | •       | •        |
| Operating System Support             |          |        |         |          |
| Windows                              | •        | •      | •       | •        |
| MacOS                                | •        | •      | •       | •        |
| Linux                                | •        | •      |         |          |
| iOS App                              | •        |        | •       |          |
| Android App                          | •        |        |         |          |

Table 3. Comparison of the functionalities and features among applications



5.2.3 *Functionalities.* These four reference management applications have very similar design of features and functionalities, but are in varied forms of presentations and work differently. The Mendeley, Zotero and RefWorks can import a reference entry by dragging the files to the interface. However, all participants experienced a longer wait time when using Mendeley. Under non-ideal connection bandwidth, the RefWorks also takes a lot longer to respond. Zotero, instead, finishes in seconds. To suit the needs of users who only use reference managers as PDF manager/reader, developers should improve the built-in PDF readers to support better labeling, commenting and highlighting tools. For nowadays's users, other important features include Google Scholar integration, team collaborations and multi-device/multi-OS support. It is never a bad idea to support these features, but the correctness and execution time should be fully considered so that new functionalities will not harm the overall efficiency.

Comparing the retrospective preference to their answers in the pre-study survey, we can also conclude that the variation in user preferences can result from unfamiliarity with other reference management applications. As reported by the participants, those who changed their minds have not got a chance to try Zotero or RefWorks. A very large portion of users are already comfortable with the current preference that achieves the same goal. As one of the participants asserted before the study, "I have been using EndNote for many years. It's not free but convenient to use." After trying between the four applications, he suggested that "Zotero and RefWorks are so good! I'd probably stop my EndNote subscription." It can be inferred that if users got a chance to compare the applications on the same task, they might reach a different conclusion. Promotion of reference management applications should let users to physically try it, either by providing tutorials for institutional subscribers or online sessions for novices—instead of merely giving out advertisements around the internet.

## 6 CONCLUSION

Reference managers are widely used in academia and research community to support the query and management of references, which is of great convenience in the development of manuscripts. This project aims to develop a way of evaluating the quality of reference management applications quantitatively to find the connection between the users' physical and mental efforts in using these applications to the users' preference for such applications, and provide suggestions to users on how to select a suitable reference manager and guidelines to developers on how to improve the current reference managers.

This work stems from the previous qualitative reviews of reference management applications and further considers the quantitative mental/physical efforts of users. The proposed method improves the traditional three-stage method (pre-study survey–participants study–post-study survey) by introducing a mental simulation and retrospective survey. The idea of evaluating the user's mental effort and change of thoughts for a specific task is not widely adopted in the comparative study of applications, and we believe this method can reveal closer to real feedback from users (participants) regarding their experience with the applications.

We also recognize the limitations of this work. For the group of subjects, although there is some diversity in their habits of using reference management applications, the group can not well represent the actual users of the applications. Senior graduate students, research professors, librarians can be expected to use such applications with a much heavier workload, for which their participation may lead to different study results. The measurement of the physical effort of users is based on a small simple task, which may not be comprehensive in revealing the applications' full functionalities or all of the users' needs. More advanced measurements should be developed to provide more insights into the applications' usability.



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